WORLD RESOURCES INSTITUTE | ROSS CENTER

# KRT

KHARTOUM-MAP

. السكة سكتنا .

## KhartouMap Initiative

**Oct 2023**

Khartoum, Sudan | Cambridge, MA

## Abstract

Sudan's national capital area consists of the three major cities of Khartoum, Khartoum North (Bahri), and Omdurman. The Greater Khartoum region and its seven localities are home to nearly 8 million people (2020 estimate), approximately a fifth of the country's total population. Most of the country's governance, public and academic institutions, trade markets, and other services are based in this Tri-City area. Sudan has one of the lowest car ownership rates in the world, standing at 2.7%. While this rate is higher in Khartoum, the vast majority of the population is transit-dependent.

Despite the transit dependence of the population and the existence of over 300 public transit routes connecting the different areas of the Greater Khartoum region, there is no official system or network map available to the public. Word-of-mouth remains the only way to get instructions to navigate Khartoum's public transport network. This situation results from years of unjust international sanctions that severely restricted access to even the most basic tools and technologies, including mapping software (e.g., ArcGIS, Google Maps, etc.) coupled with bad governance and a lack of investment in essential public services.

The KhartouMap initiative worked to methodically resolve this issue by mapping the existing system, generating GTFS-compliant data feeds that can be integrated with the now-accessible mapping services, and conducting a large-scale household mobility survey for the region. This report details the steps of work conducted by the KhartouMap team, the challenges encountered, and how we believe the data collected through this work can be utilized to analyze the status quo of mobility in Khartoum and methodically develop innovative future mobility solutions.

# Table of Contents

# Introduction

KhartouMap addresses the pressing public transit needs of one of Africa's busiest urban capitals - Khartoum, Sudan. The transit system in Khartoum is composed of an aging fleet of buses and vans that are privately operated. The lines run predominantly on government-charted service routes in the tri-city capital. This is collectively referred to as the '*Muwasalat*' system. The main vehicles in the fleet are Mitsubishi Rosa buses (capacity: 25), Toyota Hiace Vans (capacity: 12-14), and Amjad vans (capacity: 7-8). There are also a very limited number of government vehicles, colloquially named '*bus al-wali*' (capacity: 45-50) which translates to 'the governor's bus'. The system is predominantly single-modal and reliant on these bus routes and their carrying capacity. According to previous reports, approximately 80% or more of the population uses public transport - the majority of whom rely on it exclusively. There is a commuter rail system functioning at an extremely limited capacity. Though tracks exist, they are poorly maintained and train maintenance using spare parts is an ongoing challenge.

Extreme shortfalls in transit supply have crippled the capital in recent years. Coupled with inflationary pressures, soaring gas prices, and a poorly maintained registration system for bus capacity by route, there has been a chronic lack of public transit availability. The prices of rides on the system have also ballooned exponentially. This has resulted in undue hardship for public transit riders. In addition, visibility into the system is particularly opaque. Transit riders are not able to see the full system or know the fares of parallel routes, which creates a situation where imperfect information leads to a lot of inefficiency and unideal outcomes such as long queues and price gouging. All those issues have been further exacerbated by the ongoing conflict in Sudan since April 15th, 2023.

The first aim of our work at KhartouMap is to fully map the existing public transit system in Khartoum, Sudan, and create both physical and digital maps of bus routes. From these, General Transit Feed Specifications (GTFS) (i.e. digitally compliant transit feeds) are created in order to integrate the bus system information into popular mapping applications such as Google Maps and Apple Maps. The project scope also included mapping transit accessibility with special attention to geographic region, socioeconomic class, gender, and trip purpose. This has been accomplished through GIS analysis, as well as neighborhood-level mobility surveys which have been collected both in-person and online. We see KhartouMaps' mobility database as a substantial increment to previously conducted work in the Khartoum Master Plan from 2014.

## Related Work

Most countries in Africa including Sudan do not have GTFS and system maps that are available to the public to use and obtain information about transit systems. At the time of writing, only 16 African ***cities*** had a public GTFS. However, some recent initiatives have started collecting data about the transit system and developing GTFS in different countries. One of the most relevant studies to this initiative is The Digital Matatus project which was conducted in Nairobi, Kenya. A study by Williams et al. 2015 discusses this prior project. The Digital Matatus project used the geo-locative capabilities of mobile devices to map the semi-formal transit in Nairobi, Kenya. The data was collected using different smartphone applications such as MyTracks and TransitWand. The team documented some of the challenges they faced using mobile devices which included limited battery life, the slow speeds of affordable Android phones, and small screen sizes for phones for data entry. Despite these challenges, the team found that collecting data using mobile devices was the most effective approach for their purpose. The team was able to use this data to develop the GTFS and to upload it to Google Maps for riders to benefit from (Williams et al. 2015). Following the success of this project, other cities such as Accra, Addis Ababa, Cairo, Cape Town, Dhaka, Douala, Durban, Kampala, Lusaka, and Maputo to map their transit system (Klopp and Cavoli 2019, Zegras et al. 2014). These projects show that using mobile devices to map semi-formal transit is a growing trend in the global South, and they also inspired the KhartouMap project to map out and develop GTFS for semi-formal transit in Khartoum.

## Data Collection and Processing

Data collection was the main phase of the project and it is divided into two parts. The first part was transit route mapping and the second was conducting a comprehensive mobility survey in Khartoum, Sudan.

### *Transit Routes Mapping*

This part aims to collect route data to develop a map of all transit routes in Khartoum and then use them to develop Khartoum state GTFS. The first step in this part was collecting existing data from the Ministry of Infrastructure and Transportation (MOIT) in Khartoum State. MOIT provided three datasets to the team. The first dataset was a Word document that has the names of all routes that were active within the past few years. The second dataset was a spreadsheet that had a description of some of the routes, and the third was a shapefile that had the shapes of some routes that were developed about ten years ago. To handle these different datasets, the team created a workflow as shown in Figure 1. This workflow starts the *Validation Lead* of the project by selecting a certain number of routes from the inventory of routes provided by MOIT for processing. Routes are assigned to one of the *Mapping Leads*, whose first step is to go to the start or end station to verify if this is still an active route, accounting for the lack of inventory tracking by the MOIT.

If a route is no longer active, that would be documented as an inactive route in our database for the record. If the route was found to be active, then this route would be checked in the other datasets. If the route has records in both datasets (i.e. has both a route description and a shapefile), the description and the shapefile would be cross-checked by both the Validation and Mapping leads. In case the route does have a record in either of the two datasets or if the description and the shapefile do not match, this route will be designated as a route for mapping by the on-ground team (i.e. for collecting GPS track data for the route). Cross-validated routes and GPX-tracked routes are then sent for production in the final system map.

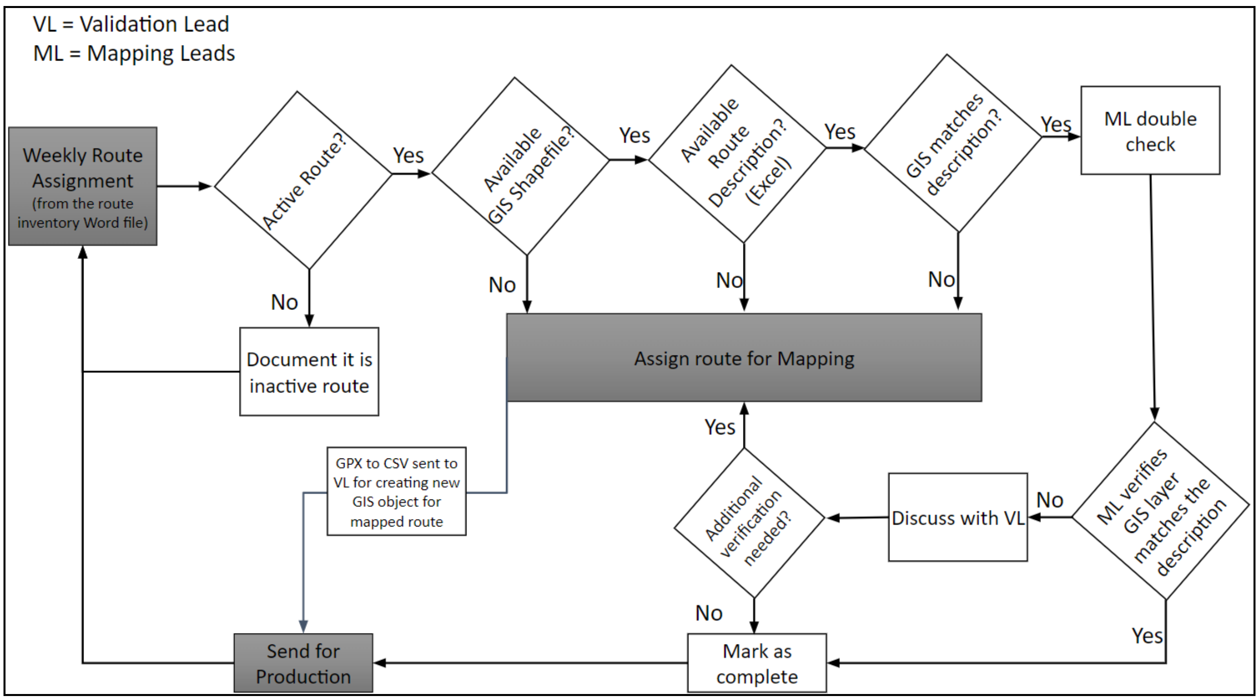


**Figure 1. Validation and mapping workflow.**

If the route was sent for mapping, then a Mapping Lead (or designated subcontractor) would ride a transit vehicle that operates on the route and collect GPS Exchange (GPX) tracks using open-source GPX tracking software. *OSMtracker* is shown in Figure 2. We developed automated corrective scripts utilizing the Valhalla open-source routing engine that map-match noisy GPS signals to align with the OSM roadway network, addressing any GPS inaccuracies as illustrated in Figure 3. Map-match route coordinates are then utilized to create transit route shapefiles. The code of map-matching GPX tracks through Valhalla is open-sourced in KhartouMap's GitHub repository.

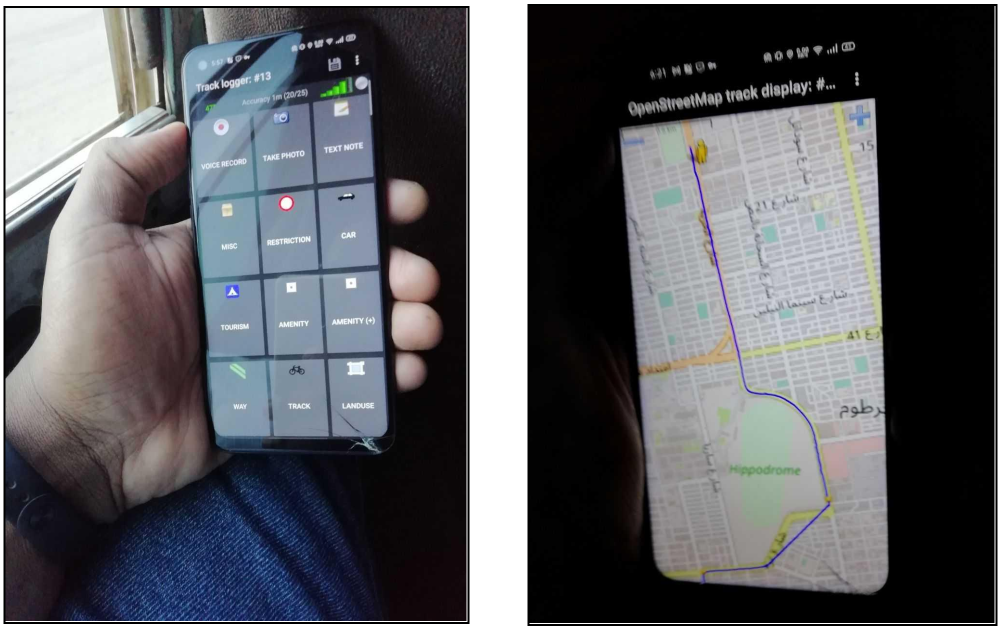


**Figure 2. GPX tracking on public transport routes.**

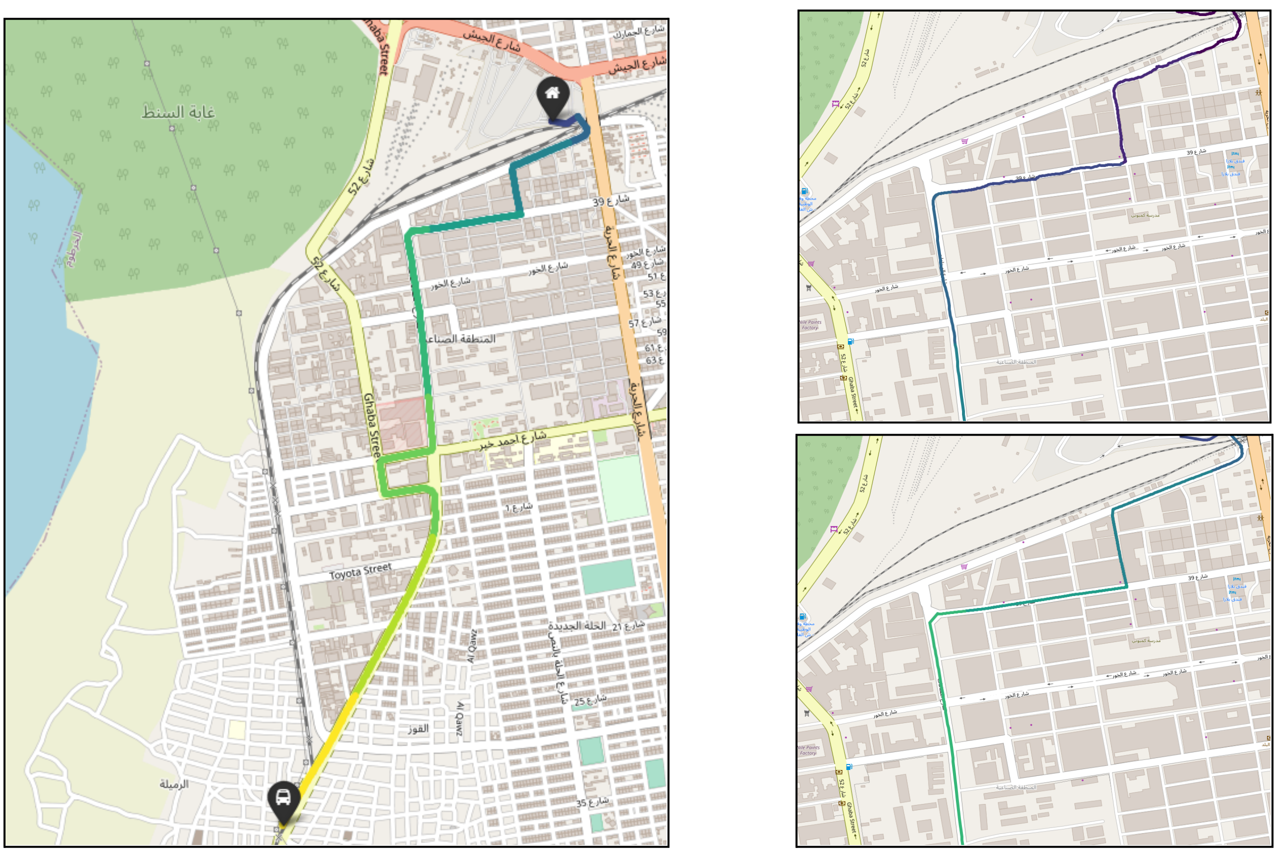


**Figure 3. Tracking (left), Original GPS Tracks (top-right), and Map-Matching (bottom-right).**

*Generating Khartoum Beta-GTFS*

Our team has mapped and validated over 300 routes serving the seven localities of the Khartoum State. Following data route data collection, we have started working with local authorities at the MOIT to generate the State's and Nation's first GTFS feed. While the progress on this, as with the rest of our project, was severely impacted by the ongoing conflict in Khartoum, our team has produced a Beta-version GTFS. The GTFS dependencies generated include the routes, shapes, times, and stops files, as well as the general information on operating hours and time-of-day service level estimates. This data has been used in our accessibility analysis covered in later sections of this report and will be open-sourced with the raw data collected by the KhartouMap team.

Figure 4 illustrates the main transit corridors and stops (generated based on route origins and destinations) for the State of Khartoum. In-route stops are not illustrated as the system operates on a continuous pick-up and drop-off basis anywhere along the transit route. The appendix of this report includes a detailed map illustrating all the routes, origins, destinations, and main terminals that serve the system.

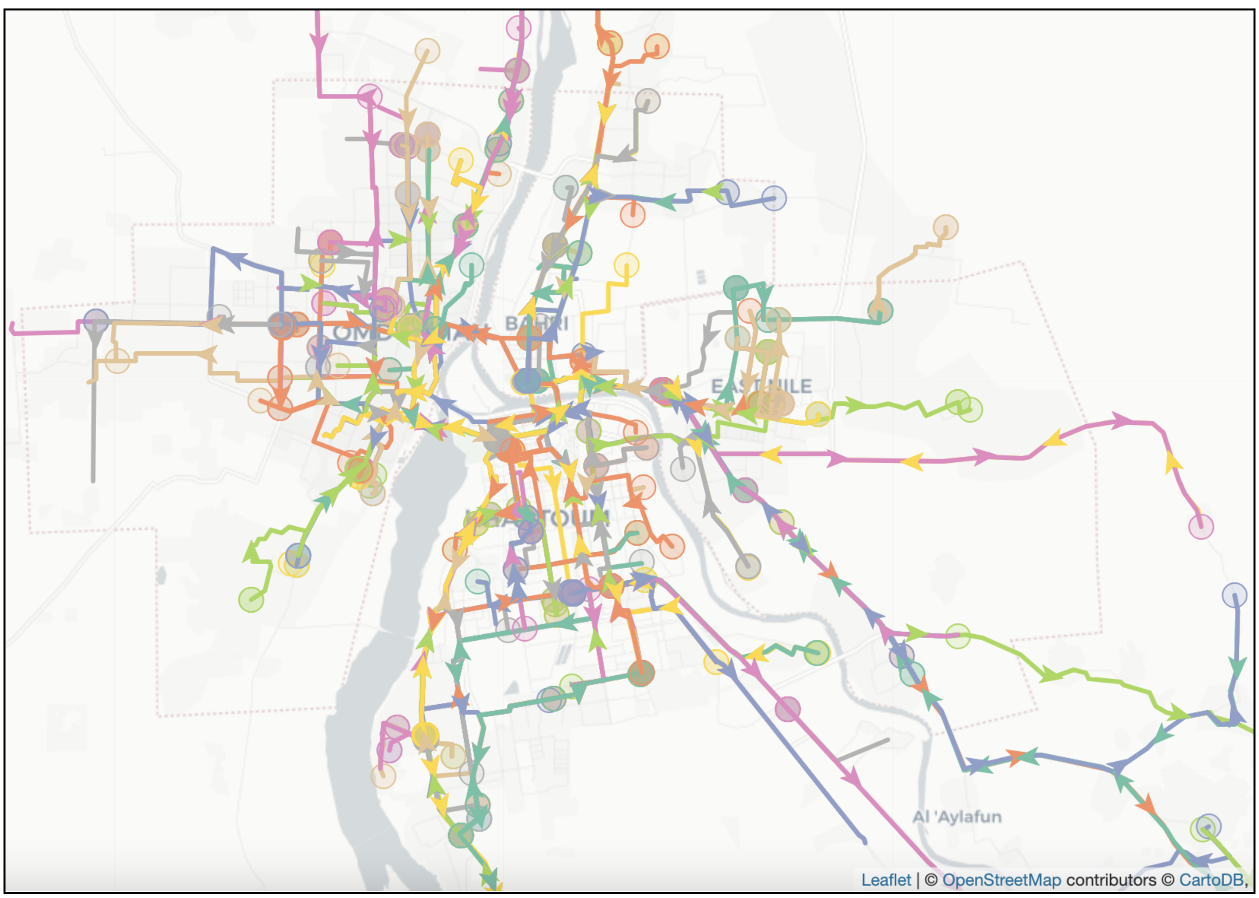


**Figure 4. Main transit corridors and stops from the Khartoum Beta-GTFS.**

### *Mobility Survey Data Collection*

To accompany the physical network map, we designed a comprehensive mobility survey to collect qualitative and quantitative information regarding the usage of public transit in Khartoum. The survey included questions about public transport frequency of use, purpose, most frequent trip origin and destination, travel time, accessibility to public transportation, and demographic information. The survey was launched online on different social media platforms (Facebook, Twitter, LinkedIn, and WhatsApp). We collected approximately 7,000 responses to the survey, a testament to the public engagement efforts that our team has fostered throughout the project. Figure 4 shows a snapshot of the responses (when we had a little over 5,000 responses) illustrating how our reach covered individuals with diverse transit use, age, employment status, and income levels.

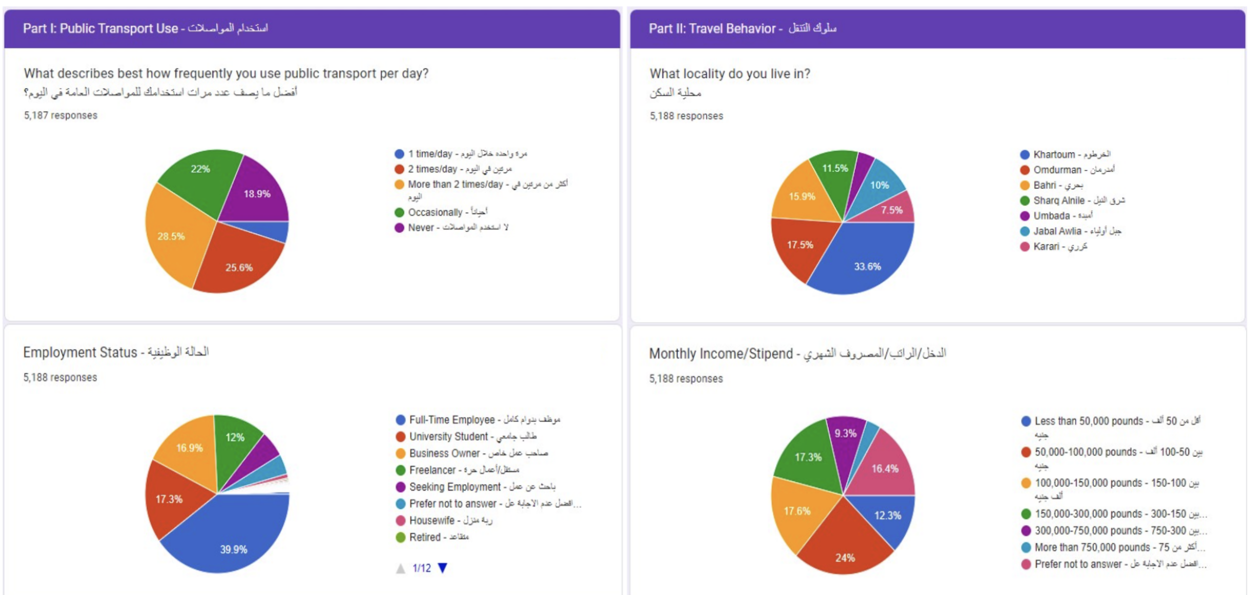


**Figure 5. Snapshot of responses from the KhartouMap mobility survey.**

In addition to the online survey, we hired on-ground contractors to collect surveys in the different bus terminals in Khartoum. We created this survey stream to reach out to individuals who might not be on social media, although the multi-generational housing arrangement of Sudan allowed us to reach different age levels through the youth in the house on social media as illustrated above. The on-ground data collection phase targeted areas (terminals and neighborhoods) that were less represented after geocoding the online survey results. This on-ground effort commenced in early April but was shortlived due to the start of the ongoing conflict in Khartoum on April 15th. We collected about 700 responses (~10% of our survey total) during the few days that this effort was sustained.

Figure 6. KhartouMap team members conducting on-ground mobility surveying.

### *Processing Survey Data*

The survey responses from both the online and on-ground streams were combined, verified for accuracy and missing data, and exported as one file. In addition to the usage and demographic information provided by the respondents in the survey, we used text matching and geocoding of self-reported origins and destinations of trips to match responses (where possible) to geolocation at the neighborhood level. We were able to geocode 89% of origins and 75% of destinations, which are illustrated in Figure 6. Additionally, the survey data open-sourced as a part of this project flags whether the response was received online or in-person to allow for studying if there were differences within those populations of respondents.

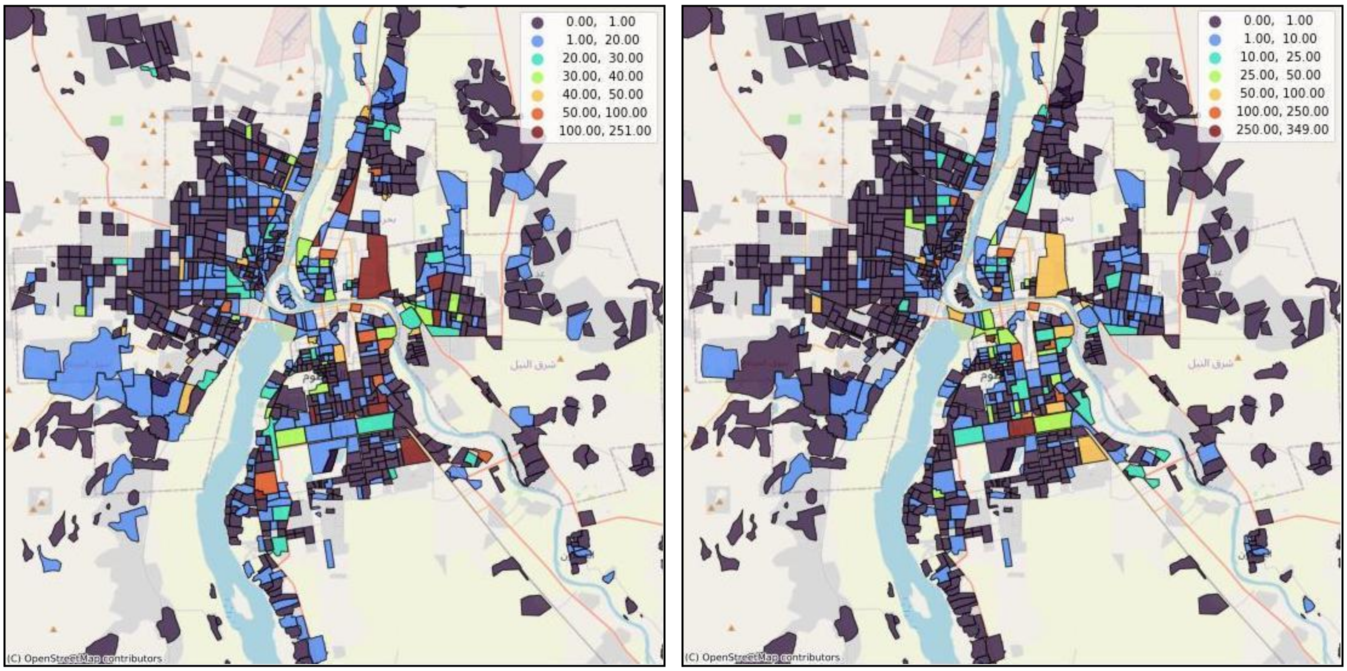


**Figure 7. Geocoded origins (left) and destinations (right) of the KhartouMap mobility survey.**

### *Analyzing Survey Results*

Analysis of survey data yielded many valuable insights into Khartoum's public transit system. Survey responses were distributed from riders across all seven localities and had a representative age range. Survey responses tended to skew males (75%). Most riders surveyed were regular public transit riders, using the system twice a day or more. The reported purposes for trips were varied and ranged from work (most common) to household grocery shopping, social trips, shopping for self, school or university, and recreation.

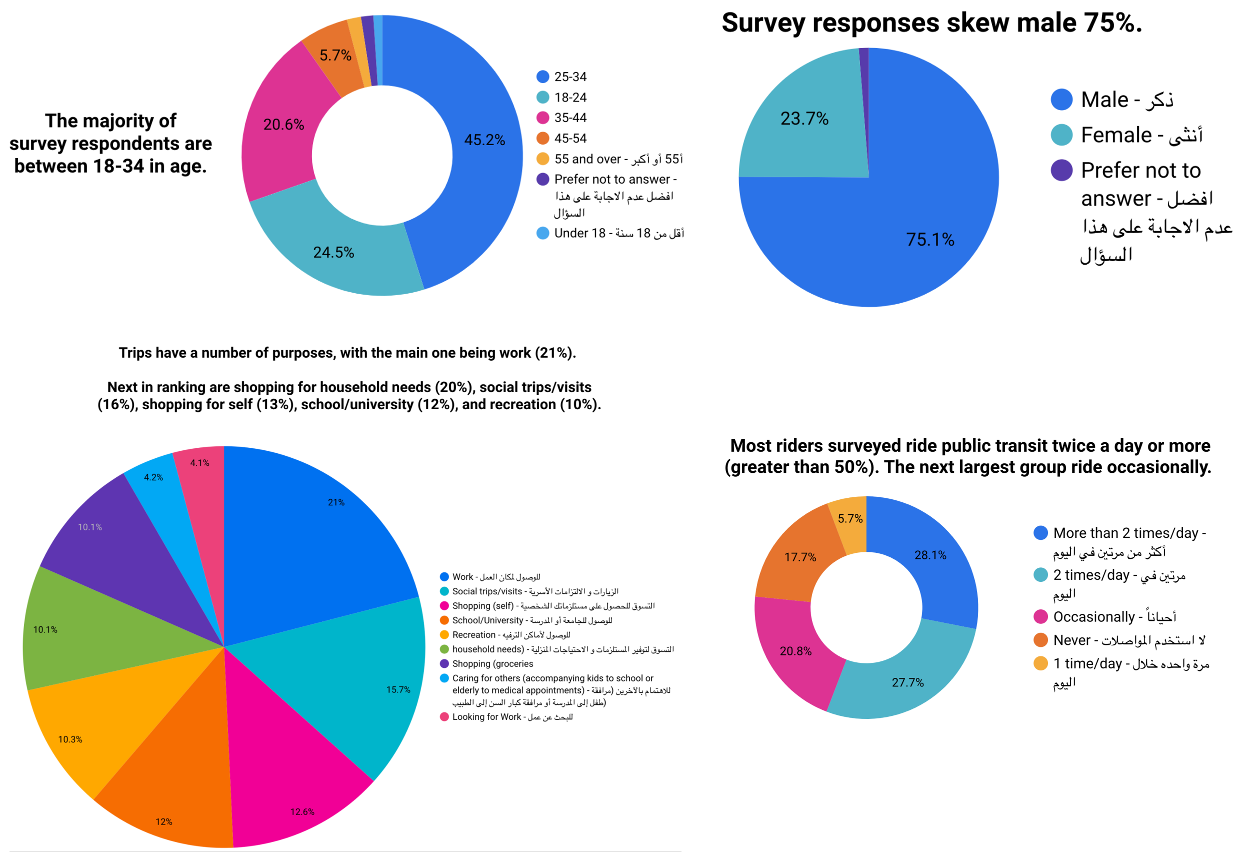


**Figure 8. Respondent demographics and ridership frequency from the KharouMap mobility survey.**

Several issues were self-reported by transit riders about to system. The top issues identified by riders were the following: (1) Overcrowding, (2) Reliability, (3) Cost, (4) Driver Behavior, (5) Lack of Info on Routes/Fares, (6) Personal Safety, (7) Availability, and (8) Sexual Harassment. The duration of most trips (over half of the data collected) was between 30 minutes and 60 minutes. Top destinations all tended to be in Khartoum, Khartoum, with over 75% of trips ending in a Khartoum destination regardless of origin locality. The top stops in Khartoum are Al-Arabi Market (major transit hub), Riyadh, and the University of Khartoum. Respondents were asked to report on their first trip of the day throughout surveying, and results showed that 80% of trips begin from 6:00 a.m. to 9:00 a.m. These are the peak hours of demand. Within this window, 7:00 a.m. to 8:00 a.m. is the most active time with the highest demand. Typically riders require 2-3 transit lines to complete their trip, transferring 1-2 times along the way.

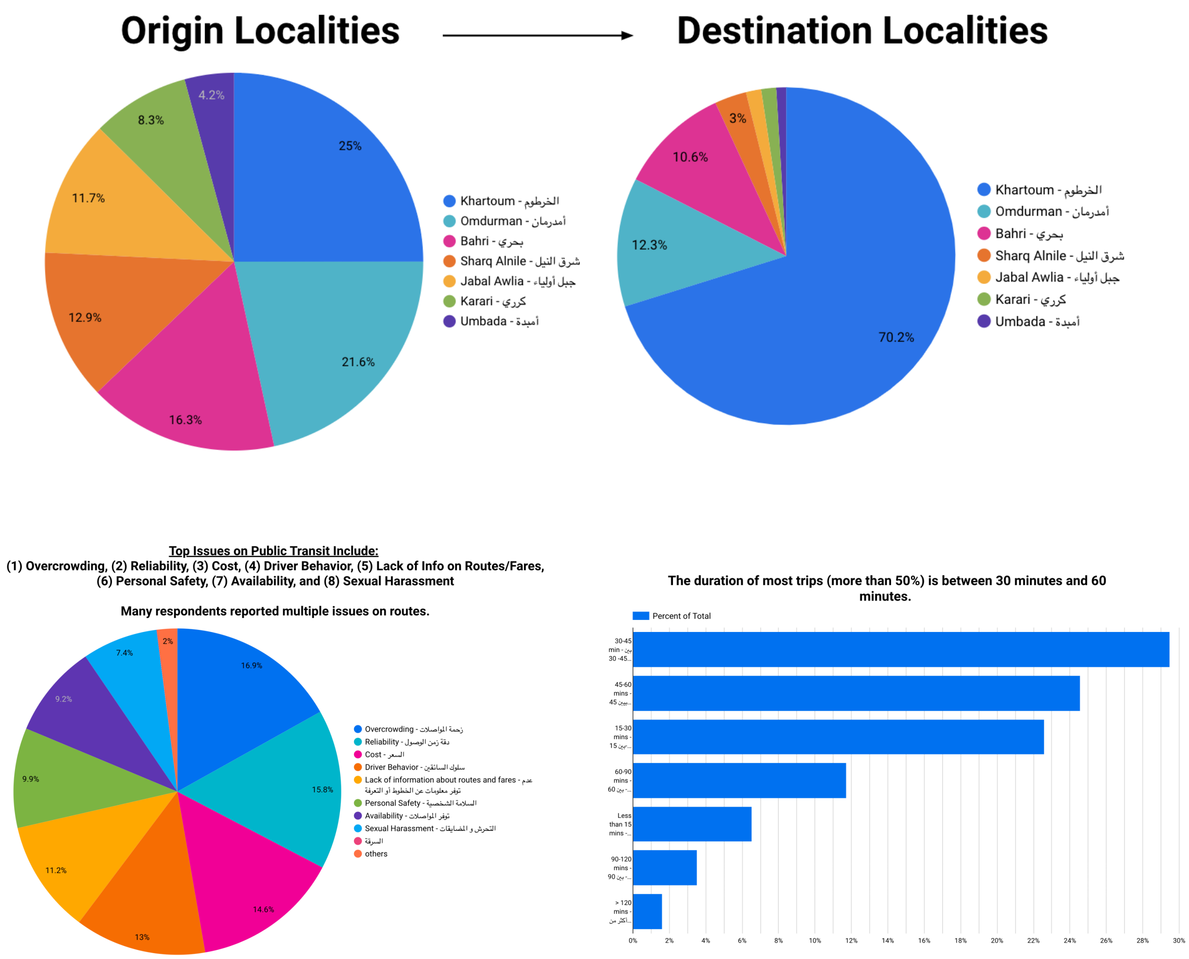


**Figure 9. Key takeaways from the KharouMap mobility survey.**

Most riders walk in order to access the public transit system (80%). A smaller portion takes a raksha (3-wheeled tuk-tuk) or ride with a friend/family to access public transit. For more than half of transit riders, this initial trip takes between 5 minutes - 20 minutes. At the time of data collection, the daily average range for spending on transit was between 640 - 740 SDG for the typical rider.

## Accessibility Analysis

One of the main purposes of the KhartouMap project was to assess service coverage and accessibility of the Khartoum 'muwasalat' system for Khartoum State. Three types of analyses were run to assess accessibility.

### *Coverage of Transit System*

First, the percent coverage of the transit network was assessed by finding the percentage of neighborhoods that are within a reasonable walking distance of the nearest transit line. According to KhartouMap surveying, 80% of riders reported walking as their main means of accessing public transit.

The best-served neighborhoods using the transit network are central areas in Khartoum, Omdurman, Jebel Awlia, Bahri, and Sharig Al-Nile. Khartoum is the only locality with 100% transit coverage within 30 minutes walking distance to all neighborhoods. Localities with large unserved areas (further than a 30-minute walk to the nearest transit line) are Bahri (34% of neighborhood areas), Sharg Al-Nile (32%), Umbada, and Omdurman (31% each), and Karrari (27%). Jebel Awlia has the second-best transit coverage at 90% transit coverage with 10% unserved neighborhood areas.

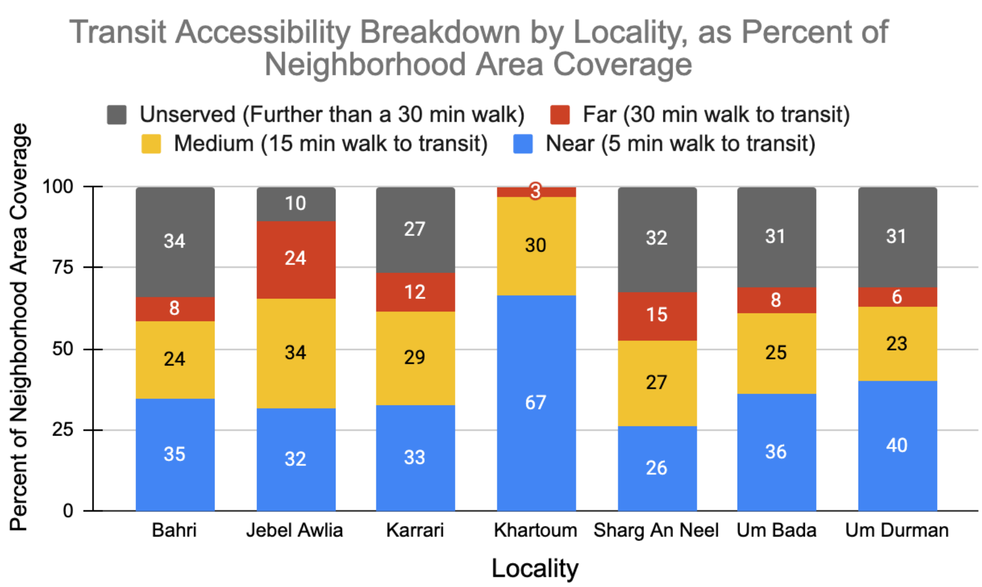


**Figure 10. Percentage of neighborhoods covered in each locality under varying access thresholds.**

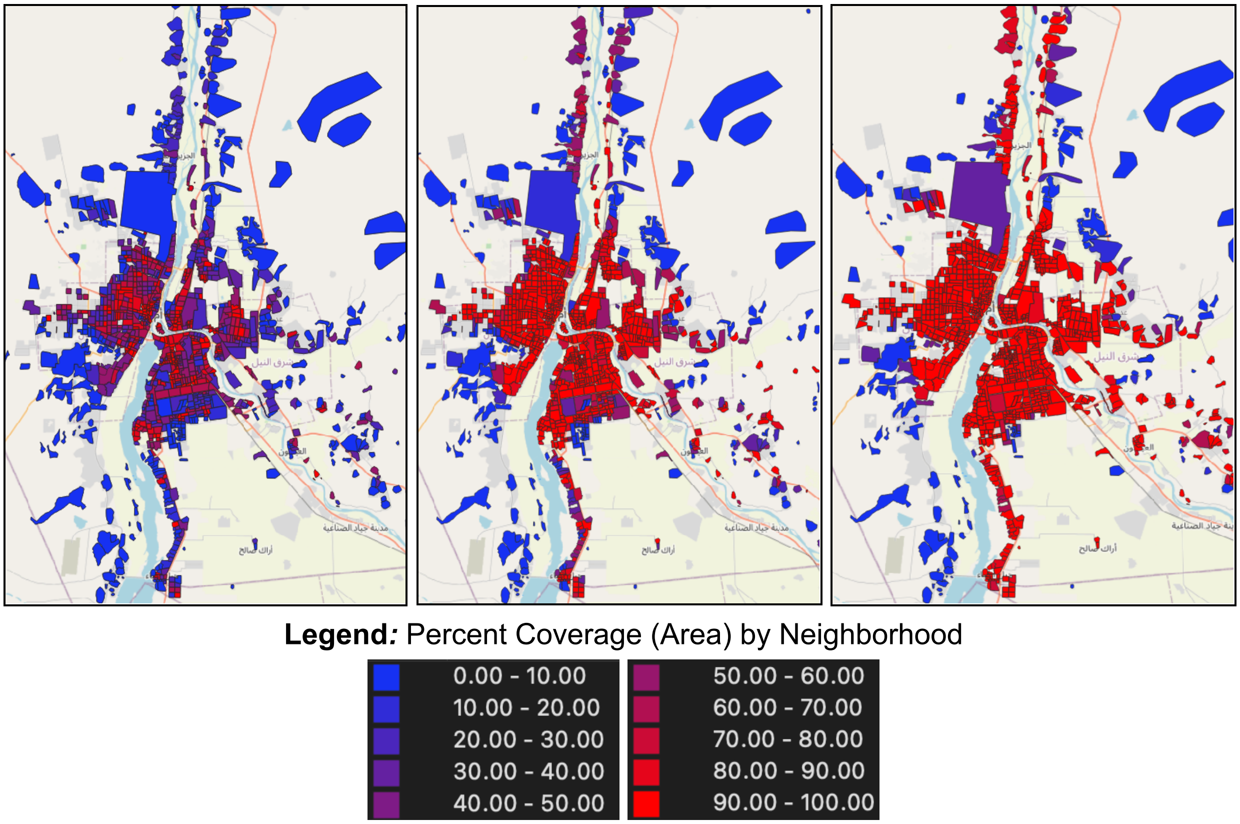


**Figure 11. Transit coverage under each access scenario (right to left: near, medium, far).**

### *Distance on Public Transit to Nearest Amenities*

Next, the accessibility of neighborhoods to the nearest amenities was also assessed. OD matrices were calculated to reveal the closest amenity for each neighborhood. The following categories were analyzed:

- Hospitals
- Banks
- Universities
- Colleges
- Schools
- Marketplaces
- Courthouses
- Bus Stops
-Supermarkets

Consistently, supermarkets and bus stops are the closest / most accessible amenities for most neighborhoods (3km on average). Next are schools (4km), banks (5km), and hospitals (8km). Distances reported in this section of analysis include the walk to and from the transit line at the entrance and exit. For the closest distance to amenities, Khartoum is the best locality (3km to all amenities on average). In Omdurman — the next best locality — the average distance is 1.5x times that of Khartoum. Each of the following localities has approximately 3x times the distances of Khartoum: Jebel Awlia, Bahri, Umbada, and Karrari. Sharg Al-Nile is generally poorly served by amenities with significantly longer distances (10x times the distances of Khartoum). Distances reported here are for the median neighborhoods of each locality to control for outliers (I.e. particularly far neighborhood areas in Khartoum State, which is a large geographic area beyond what is typically considered central Khartoum city). Please refer to the detailed accessibility analysis report to see simulated time estimates (in minutes) on public transit for the above distances.

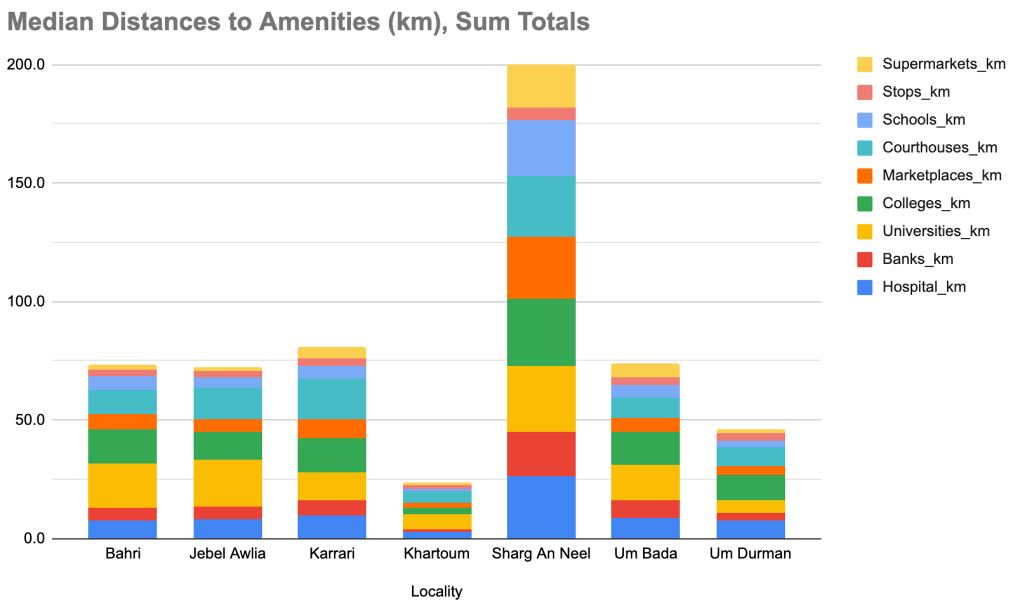


**Figure 12. Median cumulative distance to access nearest essential services by locality.**

### *Distance on Public Transit to Randomized Locations*

Lastly, from survey results, we know that the overwhelming majority of trips (70%) are directed towards Khartoum, regardless of the origin locality. In the survey, respondents were asked about their first trip of the day, so we can assume the opposite is largely true at the end of the day (mainly outbound trips from Khartoum to all localities). For this analysis section, randomized points of interest were selected in each locality, and trips were modeled to and from localities. The results are 49 unique pairs (7 x 7 matrix). Note that pairs are not bi-directional and are uni-directionally distinct (i.e. Khartoum -> Bahri is not the same as Bahri -> Khartoum). Distances reported here are only network distances, which is time traveled on the transit lines. It does not include entering or exiting the network to a final destination.

Shortest, or most efficient trips, typically originate in Khartoum going to other localities or are within a locality (such as Omdurman - Omdurman) or to the neighboring locality (such as Omdurman to Umbada). Where there is variation between a route and its inverse or opposite route, this is typically due to the greater range of neighborhood distances. Residential areas tend to have a larger disbursement in terms of geographic area than points of interest.

*Top 11 most efficient/accessible O-D pairs:*

(1) Khartoum to Khartoum (9km) - (28.1% of surveyed riders)
(2) Khartoum to Jebel Awlia (14km) - (0.3% of surveyed riders)
(3) Omdurman to Omdurman (16km) - (5.3% of surveyed riders)
(4) Jebel Awlia to Jebel Awlia (16km) - (1.0% of surveyed riders)
(5) Khartoum to Omdurman (16km) - (2.1% of surveyed riders)
(6) Umbada to Umbada (19km) - (0.5% of surveyed riders)
(7) Omdurman to Umbada (19km) - (0.3% of surveyed riders)
(8) Umbada to Omdurman (21km) - (0.8% of surveyed riders)
(9) Khartoum to Bahri (22km) - (1.7% of surveyed riders)
(10) Omdurman to Khartoum (22km) - (11.2% of surveyed riders)
(11) Khartoum to Umbada (22km) - (0.0% of surveyed riders)

Only two of the most trafficked routes (by survey results) fall within the top 10 most efficient routes through the current transit network. Those routes are Khartoum to Khartoum (28.1% of respondents) and Omdurman to Khartoum (11.2% of respondents).

*The next three heavily trafficked routes have the greatest opportunity for improvement. These routes are:*

(1) Bahri to Khartoum (34km) (10.1% surveyed riders) (#24 rank)
(2) Sharg Al-Nile to Khartoum (45km) (7.5% surveyed riders) (#37 rank)
(3) Jebel Awlia to Khartoum (12km) (7.4%% surveyed riders) (#17 rank)

*The next routes with significant room for improvement are:*

(4) Bahri to Bahri (27km) (4.7% surveyed riders) (#17 rank)
(5) Karrari to Khartoum (37km) (3.9% surveyed riders) (#28 rank)
(6) Sharg Al-Nile to Sharg Al-Nile (45km) (2.2% surveyed riders) (#38 rank)
(7) Umbada to Khartoum (32km) (2.0% surveyed riders) (#23 rank)
(8) Bahri to Omdurman (35km) (1.8% surveyed riders) (#25 rank)
(9) Sharg Al-Nile to Bahri (51km) (1.5% surveyed riders) (#42 rank)
(10) Omdurman to Bahri (30km) (1.5% surveyed riders)(#21 rank)
(11) Karrari to Omdurman (19km) (1.5% surveyed riders)

Across all routes — weighted by transit ridership percentages — the calculated average time in the transit network for the typical rider (per trip) is 35 minutes in duration and 24 kilometers in distance. This aligns with self-reported averages in the KharouMap survey where approximately 30% of riders reported a trip duration of 30-45 minutes, another 25% reported trip times of 45-60 minutes, and 23% reported trip times of 15-30 minutes.

**Table 1. Observed Origin-Destination Frequency From KhartouMap Mobility Survey**

| | | | | | | | destination_locality / Percent of Total |
|---|---|---|---|---|---|---|---|
| origin_locality | Khartoum - الخرطوم | Omdurman - أمدرمان | Bahri - بحري | Sharq Alnile - شرق النيل | Jabal Awlia - جبل أولياء | Karari - كرري | Umbada - أمبدة |
| Khartoum - ...وم | 28.06% | 2.06% | 1.73% | 0.43% | 0.32% | 0.07% | 0.03% |
| Omdurman - ... | 11.22% | 5.32% | 1.45% | 0.09% | 0.01% | 0.36% | 0.28% |
| Bahri - بحري | 10.05% | 1.75% | 4.67% | 0.2% | 0.04% | 0.01% | 0.06% |
| Sharq Alnile - ... | 7.5% | 0.64% | 1.54% | 2.17% | 0.03% | 0.04% | 0.06% |
| Jabal Awlia - ...ء | 7.44% | 0.29% | 0.36% | 0.09% | 1.03% | 0.03% | - |
| Karari - كرري | 3.89% | 1.45% | 0.58% | 0.06% | 0.01% | 0.88% | 0.07% |
| Umbada - أمبدة | 2.03% | 0.83% | 0.28% | - | - | 0.03% | 0.46% |

**Table 2. Average Nearest Distance to Randomized Cross-Locality Destinations**

| Distance, kilometers | Destination | | | | | | |
|---|---|---|---|---|---|---|---|
| Origin | Khartoum | Omdurman | Bahri | Sharg Al Neel | Jebel Awlia | Karrari | Umbada |
| Khartoum | 9 | 16 | 22 | 36 | 14 | 24 | 22 |
| Omdurman | 22 | 16 | 30 | 47 | 28 | 23 | 19 |
| Bahri | 34 | 35 | 27 | 53 | 44 | 38 | 38 |
| Sharg Al Neel | 45 | 50 | 51 | 45 | 54 | 56 | 55 |
| Jebel Awlia | 23 | 32 | 39 | 53 | 16 | 40 | 38 |
| Karrari | 37 | 29 | 37 | 58 | 46 | 23 | 29 |
| Umbada | 32 | 21 | 35 | 54 | 40 | 23 | 19 |

### *Key Takeaways*

In conclusion, large opportunities for improvement exist in expanding the transit network. One opportunity is to expand the transit network to include access to more neighborhoods, particularly in the peripheries of the State. This could include lateral expansion of the network to include smaller through-streets, which is the current norm for Khartoum, and one of the main reasons for 100% transit coverage. This would also help improve travel times for intra-locality destinations, which are both popular and in need of better service according to our analysis results. The current transit layer should be used to prioritize infrastructure planning, particularly to assess the access to amenities for different populations. It can also be used directly to assess the load on different amenities. For example, certain hospitals serve over 80 neighborhoods directly as their closest means of medical services, while other hospitals only serve 3 neighborhoods. Other factors such as specialization of hospitals, cost, and perceived quality also need to be considered, but distance and accessibility still remain primary factors. Next, careful thought should be put into major conduits of travel between and through localities. Expanding the transit network can optimize coverage and distances traveled, but speed ultimately will be capped by factors like the quality of roads, traffic, and safety (maximum speeds for buses). Khartoum is currently a single-modal transit system (primarily) using buses, but other means of transit can more effectively carry riders long distances at higher speeds and capacities through major arteries. Options include light rail, tram lines, dedicated bus lanes, and more. Future decision-makers should carefully consider these options. Lastly, another option is to connect isolated areas of the transit network to more inter-connected areas directly. This can greatly expand access for currently underserved or unserved areas with minimal lift and maximize the utility of the current system. Decision-makers should also put adequate thought into the growth of Khartoum and the surrounding areas, as plans for future growth should not be dependent only on historical information and data. Future factors like population growth, infrastructure planning, climate change, and equity are imperative to consider.

### *Limitations of Analysis*

Limitations of the current study include the inability to weigh by neighborhood population. Many neighborhoods are particularly geographically far and may not be heavily populated, but we are currently unable to include this dimension in our study. Estimates of survey route ridership provided an adequate proxy, though incomplete. Next, the KhartouMap team acknowledges the drastic changes that can be anticipated since the war in Khartoum which began on April 15th, 2023. All data presented here was collected before the war in late 2022 and early 2023, and analysis represents this time frame. It is the closest baseline to public transit coverage before the war. We hope that it will be valuable when the time comes to rebuild.

# Lessons Learned, Challenges, and Next Steps

### *Core Transit Operations*

One of the main challenges that transit agencies in other countries like Sudan face is the majority of the transit system is privately operated by individuals who own vehicles. State-operated vehicles only make up 2% of the existing fleet. Privately operated systems run based on drivers' knowledge of and experience with demand rather than on a fixed schedule, therefore it is difficult to develop fixed schedules and expect adherence. During the mapping process, our Mapping Leads and on-ground team members have noticed deviations from planned routes to be a recurring phenomenon, be it short deviations in avoidance of traffic or vehicles operating on entirely different routes than their licensed route (typically imprinted on the sides of the vehicles). Those attitudes make it increasingly difficult to provide riders with reliable information even after developing and maintaining a robust GTFS and will require additional training and oversight for compliance.

### *Public Engagement*

Our team has placed immense emphasis on public engagement to build rapport for the second phase of our project which was planned around community-centered activities. We've engaged our audience on social media channels (Facebook, Instagram, LinkedIn, TikTok, and Twitter) from the get-go, clearly communicating the goals of our initiative and the impact we plan to make. This attracted attention not only from Sudan but also plenty of expatriates and non-Sudanese from different countries around the world as illustrated below. In spite of the lack of paid social media advertising in Sudan due to sanctions, our robust network of audience grew exponentially throughout the project allowing us to reach tens of thousands of non-followers.

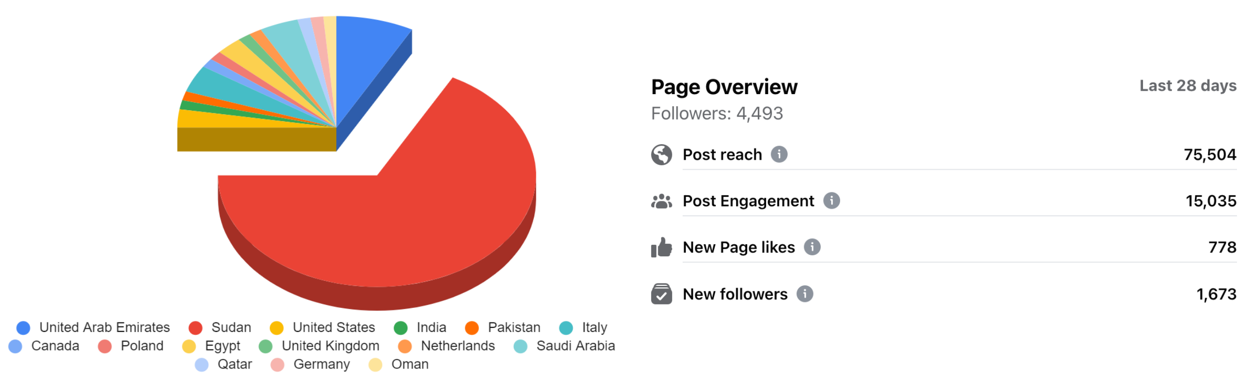


**Figure 13. Reach and engagement on KhartouMap social media.**

Although our plans for a second phase focused on community-centered activities (workshops, hackathons, and start-up competitions) were upended by the ongoing conflict in Khartoum, this robust presence on social media was incredibly helpful for us to complete our core tasks of data collection and beyond. Youth were quite excited to participate and volunteer with our team. We had over 30 routes tracked via GPX by everyday users who wanted to document their trips on public transport after a social media post by our team that explained how everyday users can support us by installing a GPX tracker and crowdsourcing their data. The value of our public engagement effort was most evident when we reached over 5,000 respondents in less than one week after launching our mobility survey, eventually receiving ~7,000 responses. While our initial hypothesis was to test the survey online and work to get the bulk of responses on the ground, our on-ground surveyors were often met with varying levels of skepticism about "potentially collecting data for government" which people aren't quite fond of. 90% of our survey responses came from online channels. Although our online audience is mostly on the younger side (35 years and younger), the multi-generation home structure of the Sudanese community allowed us to reach diverse age groups by encouraging online survey takers to ask their family members to take the survey as well.

### ***Partnerships and Working in Sudan***

In addition to building an incredible internal team that has conducted the work outlined in this whitepaper, we pride ourselves in the partnerships that KhartouMap was able to form. As we planned this project with a focus on route mapping and mobility survey data collection, we worked to initiate communication channels with the leaders at the local government in Khartoum explaining the scope of work, the benefit it will present to the state, and the country, and the means by which they can support this youth-led initiative. We've since established an excellent working partnership, which provided a safe working environment for the local contractors we hired to collect data, and allowed us to form robust partnerships with private businesses that would have otherwise been deterred from participating in such a large-scale project that would inevitably draw the attention of the government. This was in addition to existing connections with research and practice leaders at the MIT Urban Mobility, Transit, and Civic Data Design labs. As we thought of planning a follow-up phase after data collection with a bigger focus on disseminating the data, generating new knowledge, and identifying impactful use cases, we partnered with Zahara for Education, a social enterprise focused on creating and fostering educational opportunities between Sudan and experts in the United States. Through Zahara, had direct channels with local universities and entrepreneurship communities.

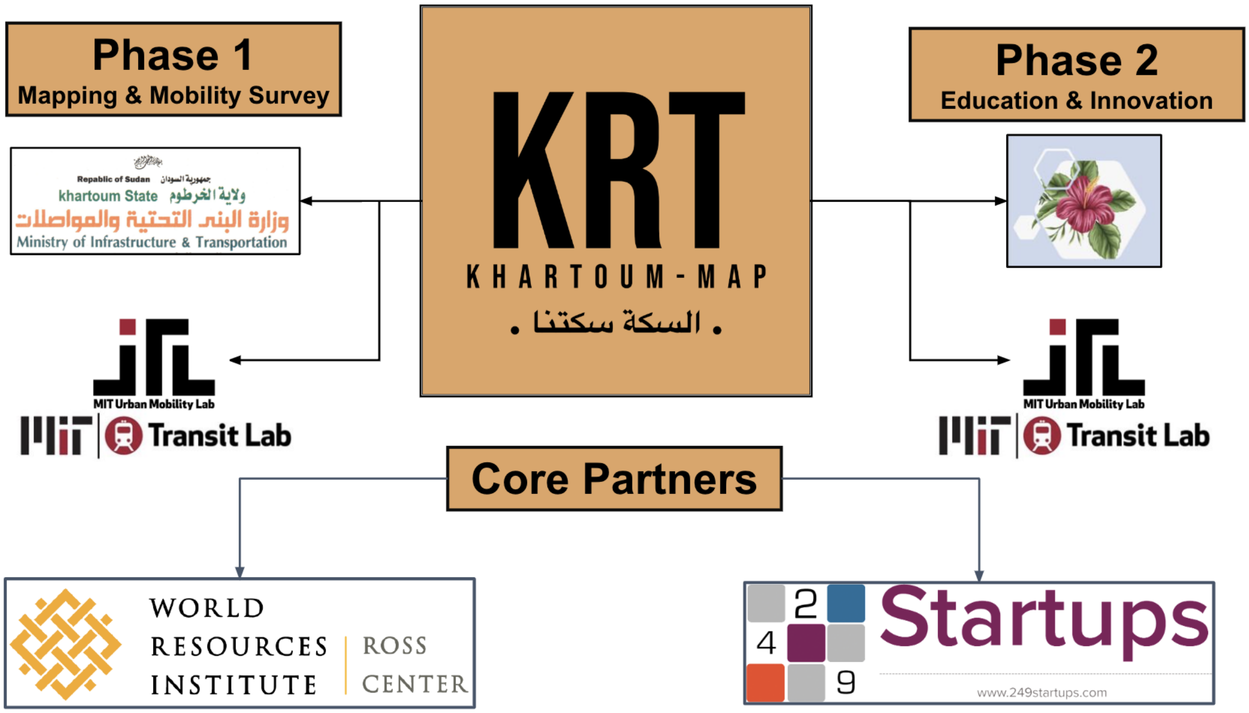


**Figure 14.Mapping KhartouMap's partnerships.**

In addition to those partnerships, our team has met with the leaders of *Transport for Cairo*, the *Association of Sudanese American Professors in America (ASAPA)*, *Savannah Innovation Labs* (a startup incubator and innovation space in Khartoum, Sudan), and *GO* (a shared mobility provider in Khartoum, Sudan). We will continue to work on forming strong partnerships with entities that will add value to the future works that will build up on this rich mobility database we produced.

While those great partnerships and immense community engagement have made navigating this project quite rewarding, the political instability - demonstrated to the extreme by the ongoing conflict in Khartoum which has completely halted our work, displaced our team members and millions of citizens, and caused widespread destruction and loss of life - is a somber reminder of the immense externalities that make work in places like Sudan challenging. This situation exacerbated the already difficult environment of funding work in a country strangled by unjust international financial sanctions, a fragile economy, and a local currency that loses value by the hour. It is in such environments, we believe, that continuing to do such work is not only necessary because it provides much-needed technical solutions but also hope. We aspire that KhartouMap is only the start of a wave of youth-led projects and initiatives that will rebuild a better Sudan.

## References


- Klopp, Jacqueline M., and Clemence Cavoli. "Mapping minibuses in Maputo and Nairobi: engaging paratransit in transportation planning in African cities." Transport Reviews 39, no. 5 (2019): 657-676
- Williams, Sarah, Adam White, Peter Waiganjo, Daniel Orwa, and Jacqueline Klopp. "The digital matatu project: Using cell phones to create an open source data for Nairobi's semi-formal bus system." Journal of Transport Geography 49 (2015): 39-51.
- Zegras, P. Christopher, Emily Eros, Kuan Butts, Elizabeth Resor, Stephen Kennedy, Albert Ching, and Muntasir Mamun. "Tracing a path to knowledge? Indicative user impacts of introducing a public transport map in Dhaka, Bangladesh." Cambridge Journal of Regions, Economy and Society 8, no. 1 (2015): 113-129.
- Central Bureau of Statistics, Khartoum State Population Estimates, 2020, http://www.cbs.gov.sd/
- MEFIT Consulting Ltd., Khartoum Transport and Mobility Master Plan. Khartoum Ministry of Infrastructure and Transport, 2014.
- United Nations Office for the Coordination of Humanitarian Affairs (OCHA) country office in Sudan, Administrative Boundaries for the State of Khartoum, 2020, https://data.humdata.org/dataset/sudan-administative-boundaries-2
- NextGIS Data, Khartoum State Basemap, 2023, https://data.nextgis.com/en/region/SD-KH/base/
- The KhartouMap Initiative, 2022, https://sd.linkedin.com/company/khartoumap